Signal Provenance, Part 2. The Structural Harvest: Why SEBI's Retail Protection Measures Do Not Reach the Losses in Indian Index Derivatives

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Abstract

SEBI's studies report about Rs 3.85 lakh crore of individual traders' net losses in Indian equity derivatives between FY22 and FY26. Participation fell after the measures introduced from October 2024, but 87.7 percent of individuals in the FY26 broker sample lost money. This paper examines what the regulatory record and aggregate exchange data can establish about costs, counterparties, positioning, trading hours and signal provenance.

The paper is Part 2 of the Signal Provenance series. Part 1 identified the missing link between a signal and the resulting order. This paper considers that gap alongside retail investor protection. It distinguishes access restrictions from surveillance and settlement reforms, including SEBI's existing responsibility rule for AI use. Contested interim orders concerning index trading and closing auctions illustrate specific alleged mechanisms; they are not treated as final findings or explanations of all retail losses.

NSE participant data shows a reversal in index-futures positioning: the Client category's October 2026 average long share was about 84 percent, compared with 9 percent for FII. Client is not a verified proxy for individuals. From 21 August to 22 October 2026, foreign investors sold a net Rs 1,06,956 crore in cash equities while domestic institutions bought Rs 1,57,928 crore and NIFTY fell 6.45 percent. A futures-only mark-to-market approximation cannot identify household losses or total portfolio returns. In a separate 19-expiry reconstruction using explicit exchange expiry identifiers, max pain was closer to the expiry-day underlying close than the previous close in 12 cases, but median misses were larger in three of four groups. This small selected sample establishes no robust forecasting advantage or manipulation mechanism.

Four proposals address information at the point of trade. A provenance category carried with the order would record a reported source category without message content. A pilot should test completeness, reliability and privacy before any mandate. The paper does not claim that provenance reduces losses or that the observed associations identify their causes.

Keywords: Signal provenance; Retail investor protection; Securities regulation; SEBI; Market microstructure; Index derivatives; Index options; Algorithmic trading; Order audit trail; Information asymmetry; Retail trading losses; Expiry-day settlement; Market surveillance; Foreign portfolio investment; NIFTY; SENSEX; India

JEL codes: G18, G14, G13, K22, G28, D82, G41

Declaration of interest. The author is the founder of AION Analytics (India), which provides market information services on Indian index derivatives and has filed provisional patent applications, one of them covering the signal-provenance method described in Gupta (2026a). This paper names no AION product and recommends no trade. Every regulatory and market figure is cited to a public source; the author-held data is described in the Appendix and is available to a regulator or reviewer on request.

Funding. No external funding. **Ethics.** No human participants; public and author-collected market data only.

Not advice. This paper is information and analysis. It does not recommend buying, selling or holding any security or contract, and contains no price target. Matters described from SEBI's interim orders are prima facie findings, contested by the parties, and no final determination is implied.

1. Introduction

India's equity derivatives segment became, in the years after 2020, one of the largest in the world by contracts traded, and individuals came to it in numbers no other market has seen. SEBI counts 7.3 lakh unique individual traders in the Equity Derivatives Segment (EDS) in FY15, 26.3 lakh in FY21 and 1.06 crore in FY25 (SEBI, 2026a). The regulator measured what happened to them, repeatedly: 89 percent lost money in FY22 (SEBI, 2023); 93 percent over FY22 to FY24, with losses above Rs 1.8 lakh crore (SEBI, 2024b); Rs 1,05,603 crore in FY25 alone (SEBI, 2025d); Rs 91,685 crore in FY26 and about Rs 3.85 lakh crore over five years (SEBI, 2026a).

SEBI's response has been sustained and careful. Beginning with a circular of 1 October 2024 (SEBI, 2024a) and continuing through 2026, it raised the minimum contract size, limited each exchange to one weekly benchmark expiry, collected option premium upfront, removed calendar-spread benefit on expiry day, added margin on short options at expiry, monitored position limits intraday, measured open interest by delta, fixed expiry days, and introduced and then revised a closing auction for the expiry settlement price (Section 3).

Participation fell after the measures; this comparison does not establish their causal effect. They did not materially reduce the share of participants who lose, and they did not reduce the average loss of those who remain. If the rules have been tightened and the loss rate has barely moved, then either the rules aim at the wrong cause or the cause is not where the rules assume it is. SEBI's measures share one premise: that losses come from too much access, by people who should not be trading, so that making trading more expensive, less frequent or larger in ticket size will reduce harm. That premise treats the loss as a property of the trader. This paper treats it as a property of the market the trader enters: its cost floor, the counterparty on the other side, the position the trader ends up in, the hours in which that position's risk moves, and the information that led the trader there.

1.1 Why this is Part 2

Signal Provenance in Retail Algorithmic Trading (Gupta, 2026a), the first paper in this series, showed that SEBI's algorithmic trading framework, mandatory for every broker from 1 April 2026, closes the order chain and cannot close the signal chain: a broker receives an instruction to trade, never what caused it, and no obligation placed on a broker can capture what the broker is never sent. That paper established the gap and showed it could be closed technically. It did not show what the gap costs.

This paper examines the regulatory implications. Its hypothesis is that missing provenance makes the sources of trading decisions harder to observe; it does not estimate the losses caused by that absence. The counterparty structure is published by SEBI in an annual study. The positioning structure is published by NSE every evening. The overnight market is public. Each is visible to the regulator in aggregate, and none of it is visible to the individual at the moment a trade is placed, because nothing connects a trade to the information that caused it. The regulator can see that individuals are crowded into one side of the market and onto one day of the week; it cannot see who put them there. The individual can see a signal on a phone; they cannot see who sent it, who is on the other side, or what the other side knows. Three of the asymmetries examined here are display problems, information that exists but is shown in the wrong place. The fourth, provenance, is a capture problem, information that is never recorded at all. The display proposals can proceed independently; provenance would add information about the reported source of the decision.

The argument proceeds as follows. Section 2 reviews the literature and places the paper in the series. Section 3 sets out the regulatory record and what SEBI's own data says it achieved. Section 4 explains why measures that price access cannot reach structure. Section 5 examines the cost floor, counterparties and positioning; Section 6 time; Section 7 expiry mechanics, including SEBI's documented cases; Section 8 provenance as the unifying gap. Section 9 documents the foreign exit of August to October 2026. Section 10 proposes changes with scope and enforcement. Section 11 states the limits of the evidence and what would falsify it.

The paper departs from the outline it began with in one respect, and says so here because it matters. The outline held that option sellers steer expiry settlement toward the "max pain" strike and that a mechanical bid lifts the index on the morning after each expiry. Tests using the data available for this paper do not establish either as a reliable rule (Section 7). Both claims are widely repeated in retail trading forums and channels. A paper about the cost of unverified signals should not repeat them because they are attractive. The structural argument does not need them, and the regulator's own orders describe something more specific.

2. Literature and the series

2.1 Individual traders

The finding that individuals lose by trading is old and robust. Individual investors who trade more earn less, and the securities they buy underperform those they sell even before costs (Odean, 1999; Barber and Odean, 2000). The closest comparator to India is Taiwan, where a complete record of trades allowed Barber, Lee, Liu and Odean (2009) to measure the aggregate:

individual investors' trading losses were economically large relative to the economy and were gained largely by institutions. Later work on Taiwanese day traders found that most lose, that many persist despite losses, and that learning from experience is slow (Barber, Lee, Liu, Odean and Zhang, 2020). SEBI's findings are the Indian counterpart and are stronger: more than three quarters of loss-makers kept trading after two consecutive loss years (SEBI, 2024b), and of traders active in every year from FY22 to FY26, 65.6 percent lost in every one of the five years (SEBI, 2026b). Five years of losses is long enough for experience to teach, if experience carried the information needed to learn from it. The persistence is consistent with this paper's argument that it does not: a trader who loses learns that they lost, not who was on the other side, what the other side knew, or where the signal that prompted the trade came from.

Options add a further layer. Lakonishok, Lee, Pearson and Poteshman (2007) classified US option open interest by investor type and showed that which class holds which side of the option market is measurable and persistent. Bryzgalova, Pavlova and Sikorskaya (2023) documented the rise of US retail options trading, concentrated in short-dated contracts, and the losses retail traders incurred. India has followed the same path at greater scale and with shorter expiries: in FY25, 70 percent of Indian index options turnover was in contracts on their expiry day (SEBI, 2026a).

2.2 Counterparties, algorithms and expiry

The microstructure literature describes the other side. Algorithmic trading improved measured liquidity (Hendershott, Jones and Menkveld, 2011), high-frequency traders act as a new class of market maker that earns the spread and manages inventory tightly (Menkveld, 2013), and continuous trading creates rents for speed that are competed for in an arms race (Budish, Cramton and Shim, 2015). In this literature the algorithmic counterparty is a liquidity provider, and the question is whether it improves prices for others. SEBI's data, which shows algorithmic proprietary and foreign entities earning 99 percent of those categories' derivatives profits while individuals lose (Section 3.2), asks the distributional question this literature mostly does not.

Option prices respond to net demand (Bollen and Whaley, 2004), and option-related hedging can pin stock prices at strikes on expiration (Ni, Pearson and Poteshman, 2005; Golez and Jackwerth, 2012), with a mechanism modelled by Avellaneda and Lipkin (2003). Indian expiration-day effects have been studied since at least Vipul (2005). A search of Crossref for this paper found no peer-reviewed study of dealer-hedging-driven pinning in Indian index options. Section 7 tests the popular version of the claim directly.

2.3 What the literature does not connect

The behavioural literature explains why individuals lose and keep trading; the microstructure literature explains how the counterparty earns; the pinning literature explains a mechanism at expiry. None of them asks what caused a given individual to take a given side, because no dataset records it. That is the gap the first paper in this series identified at the level of regulation (Gupta, 2026a), and the gap this paper argues connects the others.

2.4 The series and related work by the author

Signal Provenance (Gupta, 2026a) is the base of this paper and is summarised in Section 1.1 and Section 8. *The Great Indian Illusion* (Gupta, 2026b) documented a capital-flow asymmetry in the first half of 2026: a record foreign portfolio exit absorbed in large part by domestic institutions funded by SIP contributions, and a policy response that eased friction for foreign capital but not for domestic retail; Section 9 extends it to August to October 2026 and to derivatives. *Contextual Intelligence* (Gupta, 2026c) recorded that offshore index trading and news move Indian indices with latency, and keeps a boundary between incidence and price prediction that this paper also keeps; Section 6 uses its framing. *The Signature Trap* (Gupta, 2026d) supplied the method of treating the trust assumption, not the visible artefact, as the unit of analysis. *The Uncodified Margin* (Gupta, 2026e) argued that codified data misses the contextual knowledge experienced participants use. *Asymmetric Multipolarity* (Gupta, 2026f) described the decoupling of financial from real-economy power, the macro context of the 2026 foreign exit. *Beyond the Mirror* (Gupta, 2025) showed conversational AI systems drifting from their safeguards under sustained framing, which bears on AI-generated signals.

3. The regulatory record

3.1 What SEBI measured

Year	Net loss of individuals (Rs crore)	Share who lost	Individual traders in EDS (lakh)
FY22	40,824	90.2% (89% in SEBI, 2023)	51.4
FY23	65,747	91.7%	67.0
FY24	74,812	91.1%	95.6
FY25	1,11,788 (first reported as 1,05,603)	90.9%	106.2
FY26	91,685	87.7%	87.5
FY22 to FY26	about 3,84,856		

Table 1. SEBI's measurements of individual traders' net losses in the Equity Derivatives Segment (EDS: equity futures and options on NSE and BSE). FY22 to FY24 losses from SEBI (2025d); FY25 as revised and FY26 from SEBI (2026a), which raised the broker sample from 13 to 15 and revised FY25 from Rs 1,05,603 crore. The five-year total uses the revised FY25 figure and matches SEBI's stated "about Rs 3.85 lakh crore". All loss figures are from broker samples; SEBI estimates full-population losses at about 1.11 times the sample. Trader counts are all individuals in the segment (SEBI, 2026a).

The studies agree on the shape of the problem. Transaction costs were 28 percent of loss-makers' net losses in FY22 (SEBI, 2023) and 35 percent of their gross losses in FY26 (SEBI, 2026a). Individuals paid about Rs 24,800 crore in costs in FY26; on SEBI's broker sample of 78.6 lakh active traders that is an average of Rs 31,628 per trader, against Rs 26,027 the year before (SEBI, 2026a). (The sample count, not the all-segment count of 87.5 lakh, is the basis of the per-trader

figure.) Traders under 30 were 43 percent of the total in FY26, against 31 percent in FY22, and 89 percent of them lost. Losses persist: of traders active in every year from FY22 to FY26, "only 0.5% were profitable in all five years, while 65.6% incurred losses in every year" (SEBI, 2026b). Losses fall on those with least behind them: of 122.6 lakh traders across FY25 and FY26, about 35 percent held no equity portfolio at the end of FY26, and the 78 percent with equity portfolios below Rs 1 lakh contributed 51 percent of turnover and 70 percent of losses over the two years (SEBI, 2026a).

3.2 Who earns

SEBI's studies also report the other side. In FY24, before costs, proprietary traders earned about Rs 33,000 crore and foreign portfolio investors about Rs 28,000 crore in equity derivatives, while individuals and others lost over Rs 61,000 crore; algorithmic entities produced 97 percent of FPI profits and 96 percent of proprietary profits (SEBI, 2024b). In FY26, before costs, proprietary traders earned about Rs 44,000 crore, FPIs Rs 14,000 crore, corporates Rs 8,000 crore, mutual funds Rs 3,000 crore and partnership firms Rs 3,000 crore, while individuals lost about Rs 72,000 crore; "99% of profits for FPIs and Proprietary traders were made by 'Algo entities'", and the ten largest proprietary entities earned about 75 percent of the proprietary category's gross profit (SEBI, 2026a). Proprietary traders account for about 60 percent of turnover in the segment (SEBI, 2026a).

This is the regulator's own account, and it is the most important single fact in this paper. The losses of individuals are not dissipated. Gross of costs, they are earned, in the main, by a small number of algorithmic proprietary and foreign entities.

The figures document a transfer, not its mechanism. That one category earns what another loses does not show that the earning causes the losing, or that any participant acted improperly; market-making and hedging also produce transfers of this shape. The paper's arguments about mechanism are in Section 7, which rests on the regulator's own orders, and Section 8, which concerns what nobody can currently observe. The counterparty data establishes only who is on the other side, in aggregate, which is the fact the individual does not see.

3.3 What SEBI did

- **1 October 2024.** Circular SEBI/HO/MRD/TPD-1/P/CIR/2024/132 (SEBI, 2024a): upfront collection of option premium; removal of calendar-spread treatment on expiry day; intraday monitoring of position limits; minimum contract value raised from Rs 5 to 10 lakh to Rs 15 to 20 lakh; one weekly benchmark index expiry per exchange from 20 November 2024; and an additional 2 percent extreme-loss margin on short options on expiry day. The circular's stated reason for the weekly-expiry measure was that "Expiry day trading in index options, at a time when option premium are low, is largely speculative."
- **22 October 2024.** Circular SEBI/HO/MIRSD/MIRSD-PoD-1/P/CIR/2024/143 (SEBI, 2024c) barred regulated entities from associating with persons who give unregistered advice or make claims of returns.
- **4 February 2025.** Circular SEBI/HO/MIRSD/MIRSD-PoD/P/CIR/2025/0000013 (SEBI, 2025a) set the retail algorithmic trading framework, mandatory for all brokers from 1 April 2026 after extensions (SEBI, 2025f).

- **26 May 2025.** Circular SEBI/HO/MRD/MRD-TPD-1/P/CIR/2025/76 (SEBI, 2025b) limited each exchange's expiries to Tuesday or Thursday. NSE's weekly NIFTY contract expires on Tuesday and BSE's weekly SENSEX contract on Thursday.
- **29 May 2025.** Circular SEBI/HO/MRD/TPD-1/P/CIR/2025/79 (SEBI, 2025c) measured open interest on a delta-adjusted basis, set entity limits in index options and added a pre-open session for futures.
- **1 September 2025.** Circular SEBI/HO/MRD/TPD/CIR/P/2025/122 (SEBI, 2025e) set intraday limits of Rs 5,000 crore net and Rs 10,000 crore gross delta-equivalent per entity in index options, citing "observed instances of outsized intraday FutEq positions created by certain entities in index options on the day of contract expiry".
- **16 January 2026.** A circular introduced a closing auction session in the cash market, effective 3 August 2026, and based the expiry settlement price of index derivatives on the auction closing price (SEBI, 2026c).
- **15 October 2026.** After an interim order on alleged manipulation during the auction on a SENSEX expiry (Section 7.4), a circular reverted the expiry settlement price to an index value from the volume-weighted average price of the last 30 minutes of continuous trading, and stopped trading in expiring index contracts at 15:30, both effective 7 December 2026 (SEBI, 2026d).

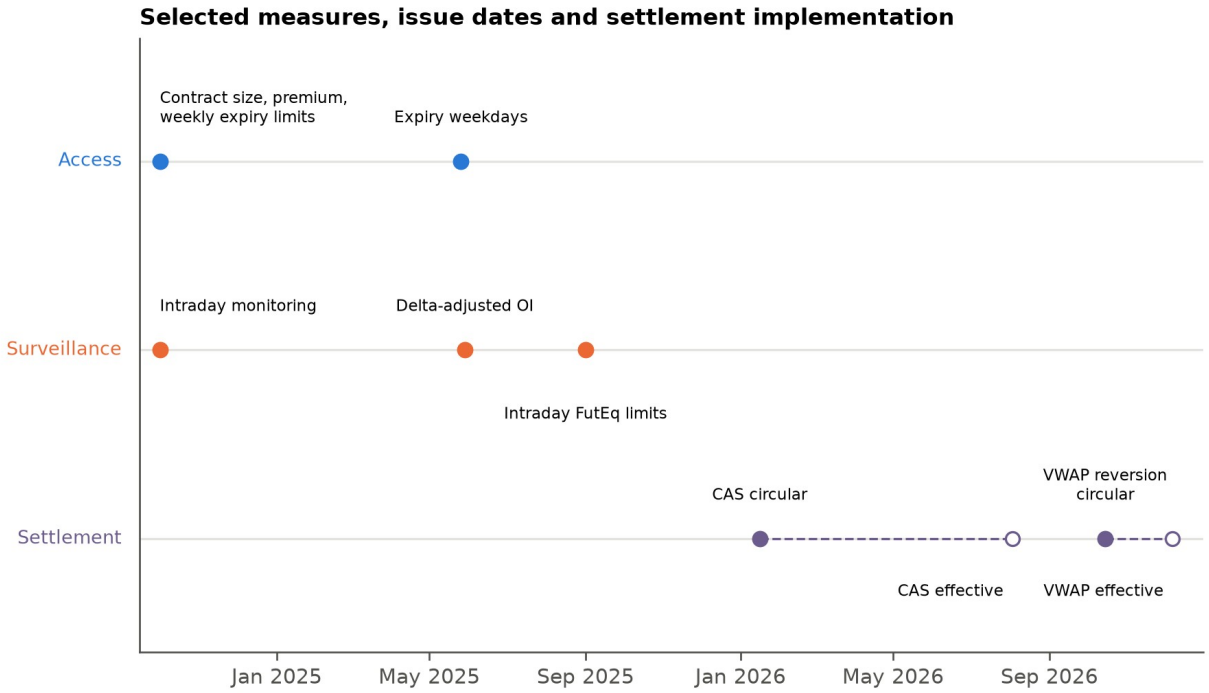


Figure 1. Selected SEBI measures by the author's categories, October 2024 to December 2026. Filled dots mark circular issue dates; open dots mark settlement implementation dates, including the announced December change. A circular can span categories. This selection shows the measures discussed in Section 3, not the effectiveness of each measure or a census of all regulation. Sources: SEBI (2024a, 2025b, 2025c, 2025e, 2026c, 2026d).

3.4 What the measures achieved, on SEBI's own evidence

The August 2026 study is the first to cover a full year under the measures. The number of individual traders fell 18 percent, from 106.2 lakh to 87.5 lakh, "the first year-on-year decline in the trader base since FY16"; new entrants fell about 40 percent; the share of index options turnover in expiry-day contracts fell from 70 to 59 percent; and aggregate losses fell (SEBI, 2026a). The share of loss-makers moved from 90.9 to 87.7 percent, the average loss per trader rose from Rs 1.13 lakh to Rs 1.17 lakh, and the average transaction cost per trader rose by more than a fifth. SEBI itself cautions that the decline in participation "does not establish a causal relationship" with the measures (SEBI, 2026a).

Read together, these figures describe a market that admits fewer people and treats those it admits about as before. That is the pattern to expect if losses are a property of structure rather than of access.

There is an alternative reading of the FY26 figures, and it should be stated. The measures may have removed the smallest and least capitalised traders, who were the most likely to lose, leaving a smaller population of larger-ticket traders. That would lower the share of loss-makers a little and raise the average loss per trader, which is what happened. On this reading the measures did help the marginal trader, by keeping them out. But the reading does not rescue the premise that access is the cause, because for those who remain the loss rate is still close to nine in ten and the average loss is higher. Both readings agree that whoever is in the market faces the same structure.

4. Why measures that price access cannot reach structure

Each measure in Section 3.3 acts on one of three things: the cost of entering a position (contract size, upfront premium, margins), the number of opportunities to enter one (fewer weekly expiries, fixed expiry days), or the scale of positions (limits and monitoring, aimed largely at the largest entities). The first two are levers on access. The third constrains counterparties, which is valuable, but does not change the cost floor, the side of the market an individual ends up on, when the risk of that side moves, or what led the individual there.

The distinction is the one *The Signature Trap* drew between a signature and a trust assumption (Gupta, 2026d). A signature-based defence matches observed artefacts against a list and reports nothing when the threat takes a form nobody has described. A behaviour-based market rule does the same: it acts on what the trader visibly does (trades too often, buys too cheaply, holds too small a contract) and is silent on the assumptions the trader relies on. The retail trader assumes, without being told otherwise, that the market visible between 09:15 and 15:30 is where the risk lives; that the counterparty to a cheap option is a neutral market rather than, in the main, an algorithmic proprietary desk; that the index used to price an option reflects the market rather than, on some days, a participant's intervention; and that a signal arriving on a phone came from someone accountable. Each assumption is false in a way the rules do not address. Experienced participants carry the corrections as uncodified knowledge (Gupta, 2026e); the first-time trader, who is now the typical trader, does not.

Raising the cost of access removes the smallest participants, leaves the larger ones, and raises the stake of each remaining trade. If the loss rate is a property of structure, it should stay close to stable while aggregate losses move with participation. Table 1 and Section 3.4 show that pattern.

5. The cost floor, counterparties and positioning

5.1 The negative-sum floor

Derivatives are zero-sum between participants before costs, and negative-sum after them. SEBI's FY26 figures show both halves. Before costs, individuals' gross trading loss was about Rs 72,000 crore, and the gross profits of the other categories SEBI reports (proprietary Rs 44,000 crore, FPIs Rs 14,000 crore, corporates Rs 8,000 crore, mutual funds and partnership firms Rs 3,000 crore each) sum to about the same amount (SEBI, 2026a). On top of that transfer, individuals paid about Rs 24,800 crore in transaction costs, of which brokerage was 44 percent, securities transaction tax 27 percent, exchange charges 16 percent, GST 11 percent, and stamp duty and SEBI fees 1 percent each. Over FY22 to FY26, individuals' costs came to about Rs 1 lakh crore (SEBI, 2026a). The gross and cost figures are on different bases (category-wise figures for the whole market, costs on the broker sample) and are not added here to a single total. Figure 2 shows the two layers.

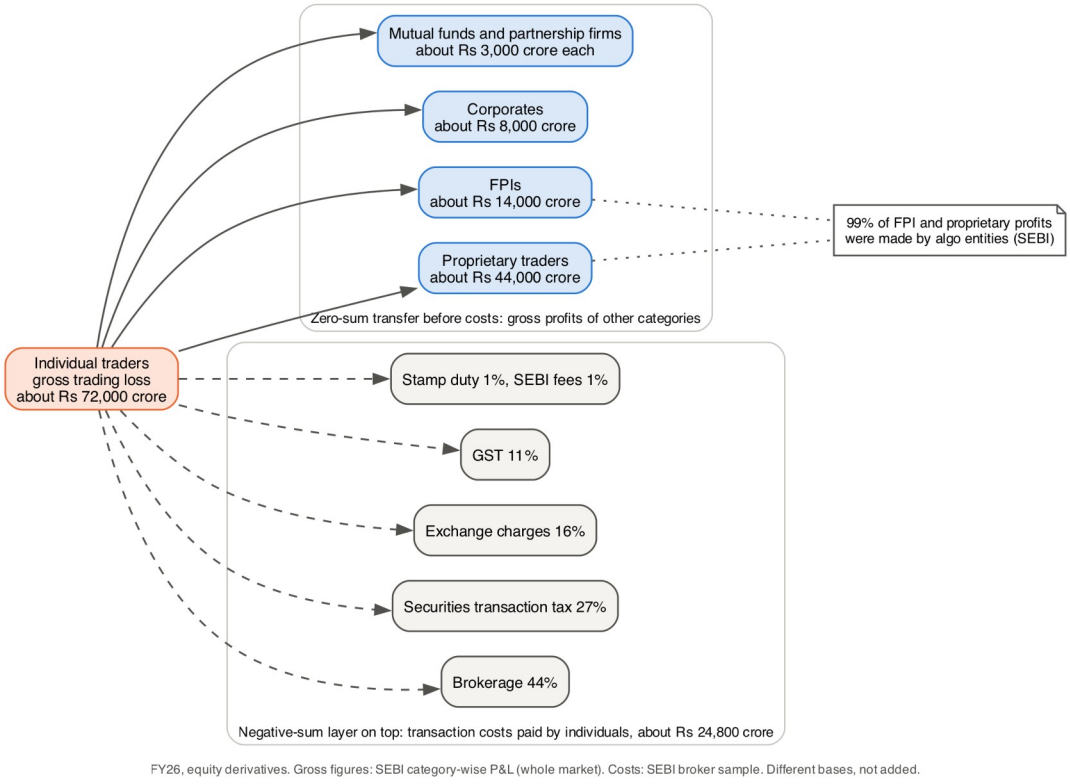


Figure 2. The negative-sum floor, FY26. Individuals' gross trading loss is matched by the gross profits of other categories (zero-sum before costs); transaction costs paid by individuals sit on top (negative-sum after costs). Source: SEBI (2026a).

The consequence is a floor under losses that no change in individual behaviour short of not trading can remove. If individuals' trades had broken even before costs in FY26, individuals would still have lost about Rs 24,800 crore. Costs also change who loses: 82.1 percent of individuals lost before costs and 87.7 percent after, because costs "pushed some traders with small gross profits into net losses" (SEBI, 2026a). For the typical trader, who buys short-dated options in small size, costs are a larger share of each trade than for an algorithmic participant trading at scale with institutional pricing, so the floor is not level. SEBI's own measures raised it: the average cost per trader rose more than a fifth in FY26, as securities transaction tax paid by individuals rose 35 percent (SEBI, 2026a).

5.2 Two channels of loss

SEBI's trading-behaviour study finds that 97 percent of individual traders were mainly option buyers and about 2 percent mainly option sellers (SEBI, 2026b). Individuals were 38 percent of index options turnover in FY26 (SEBI, 2026a), and most index options turnover is in contracts within a day of expiry. The typical individual, then, buys short-dated options and closes them the same day. The option is sold, in the main, by the participants who earn the profits in Section 3.2. That is the first channel: a high-frequency transfer, intraday and concentrated on expiry days, from many small option buyers to a few algorithmic sellers, with costs on top.

The second channel is visible in positions carried from one day to the next, which is what open interest records. Option buyers who close the same day leave little open interest. FPIs, by SEBI's account, are only about 7 percent of turnover but "around one-quarter to one-third of daily open interest" (SEBI, 2026a); their carried positions are in large part hedges of cash portfolios. The question for this channel is who holds the other side of those hedges.

5.3 Who holds the other side of the hedges

NSE publishes, every trading day, a participant-wise open interest file that reports, for each of four categories (Client, DII, FII and Pro), the long and the short open positions, in number of contracts, in index futures, index calls, index puts and the stock equivalents. Because it reports long and short separately, each category's gross and net positions can be read directly. The author has collected the file daily since September 2020; 1,502 daily files per category are used. The four categories' net positions sum to zero each day, which the data confirms.

Figure 3 shows the long share of index-futures open interest (long contracts as a share of long plus short) for the Client and FII categories by month.

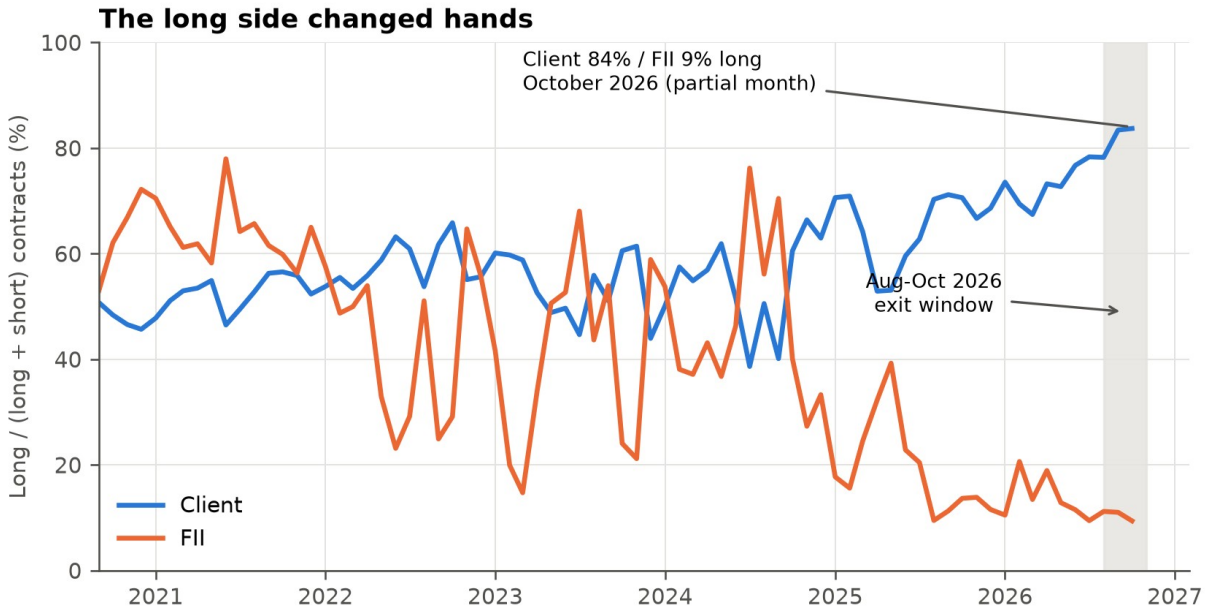


Figure 3. Long share = long / (long + short) contracts, monthly averages, September 2020 to October 2026 (partial endpoint months). Source: NSE participant-wise open interest, aggregated by the author. The chart shows who holds the long side; it does not establish why.

Year	Client long share (%)	FII long share (%)
2020 (Sep to Dec)	47.3	65.5
2021	52.5	64.1
2022	57.8	43.2
2023	53.9	40.5
2024	54.2	46.9
2025	65.2	19.3
2026 (to 21 Oct)	75.5	12.9

Table 2. Long share of index-futures open interest by year. Source: NSE participant-wise open interest; author's aggregation.

In 2020 the foreign category held most of its index-futures exposure long and the Client category was close to balanced. By 2026 this had reversed. Over the two months from 21 August to 21 October 2026 (41 daily files), the averages were:

Category	Long share of index futures (%)	Net index futures	Index puts short	Index puts long	Net index puts	Net index calls
Client	82.8	+221,912	3,246,595	2,504,729	-741,866	+252,325
FII	10.3	-268,914	508,155	1,112,930	+604,775	-295,855
Pro	63.7	+25,024	891,249	972,877	+81,629	+24,852
DII	66.8	+21,979	222	55,685	+55,463	+18,678
Total	50.0	0	4,646,222	4,646,222	0	0

Table 3. Average positioning, 21 August to 21 October 2026, in contracts. Negative is net short. Source: NSE participant-wise open interest; author's aggregation.

The Client category held, on average, 3.25 million of the 4.65 million short index-put contracts outstanding, or 69.9 percent of all short index-put open interest. It also held 2.50 million long puts, so its net position was short 0.74 million. The foreign category was the largest net holder of puts. The Client category was net long the index through futures, net long again through calls, and net short puts, that is, net long a third time by selling insurance against a fall; the foreign category was net short the index through futures and calls and held the insurance.

Three cautions apply. First, this is not evidence of wrongdoing by anyone. A foreign investor with a large cash portfolio has a legitimate reason to hedge it with short futures and long puts, and the hedge has to be bought from someone.

Second, Client is an exchange reporting category, not a measure of individual traders. No authoritative definition or individual share within it was established for this paper. Over the window it held 12.27 million of the 19.50 million index-option positions counted on both sides, or 62.9 percent. SEBI separately reports that individuals account for 38 percent of index options turnover. Turnover is not open interest, and the two classifications cannot be joined to identify individuals' positions or losses. The paper therefore makes no allocation of Client positioning or estimated mark-to-market to households.

Third, the cost and counterparty findings rest on SEBI's individual-level studies. The proposed connection between households and the other side of foreign hedges is a hypothesis requiring disaggregated data, not a finding of the participant file. The provenance proposal does not depend on that hypothesis being confirmed.

What the data shows is who the counterparty to foreign hedging has become: overwhelmingly the Client category, not the proprietary or domestic institutional categories. When the foreign category also sells its cash holdings, as it did in August to October 2026 (Section 9), the hedge pays the foreign investor and the counterparty pays for it. Option prices reflect which side carries net demand (Bollen and Whaley, 2004), which is why who sits on which side is not a detail.

Nothing in SEBI's measures addresses this. Larger contracts make each Client position larger. Fewer expiries concentrate positioning into fewer dates. Upfront premium collection bears on option buyers, while the Client category's largest carried exposure in this period was as a net option writer. The file that shows all of this is published daily, as a CSV on an exchange archive, and is not shown to anyone at the moment an order is placed.

6. Time

Trading hours changed during the sample. The domestic equity derivatives session ran from 09:15 to 15:30 IST before the closing-auction reform; SEBI's January 2026 circular specifies a 15:40 end with the reform effective 3 August (SEBI, 2026c). From 12 October 2026 GIFT Nifty index futures trade from 05:15 to 15:40 and 16:05 to 02:45 the next day (NSE IX, 2026). The earlier published 06:30 and 16:35 starts therefore do not describe the end of this paper's window. Figure 4 distinguishes current published hours from the fixed cutoff used in the calculation.

Resident individuals cannot remit margin through the RBI's Liberalised Remittance Scheme (RBI, 2024, paragraph 5). This establishes a funding restriction under LRS, not a blanket prohibition on every resident's possible means of access. Eligible offshore participants can adjust exposure during hours when the domestic exchange is closed.

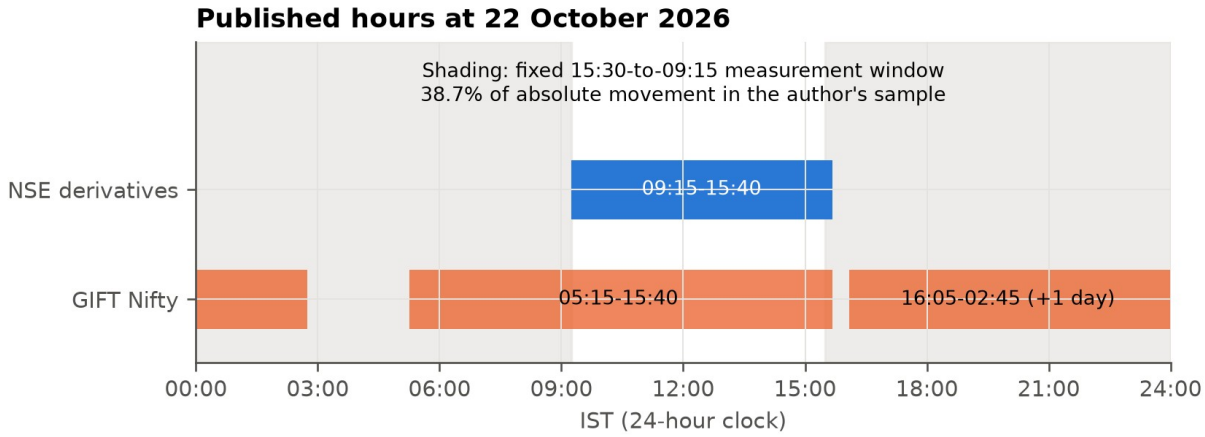


Figure 4. Published trading hours at 22 October 2026, with the author's fixed 15:30-to-09:15 measurement window shaded. The 38.7 percent is a descriptive share of absolute movement, not a variance decomposition. The window includes ten domestic trading minutes after the August reform and is not wholly an offshore-only interval. Sources: SEBI (2026c); NSE IX (2026); Appendix A.6.

Prices can incorporate information while the domestic market is closed; the data here does not identify which venue causes the next opening price. In the author's tick record of the NIFTY near-month future, covering 55 pairs of consecutive sessions between June and October 2026 (with a gap from late July to mid September; Appendix), the overnight share of absolute movement, defined as the sum over sessions of the absolute overnight gap divided by the sum of the absolute overnight gap and the absolute open-to-close move, was 38.7 percent. The median absolute overnight gap was 30.6 basis points and the median absolute open-to-close move 37.0 basis points; because the share is a ratio of sums, not of medians, the two medians do not reproduce it. Eleven of the 55 gaps exceeded 0.5 percent. The measure is a share of absolute movement, not a variance decomposition, and the sample is short; it is reported for order of magnitude, not as a constant.

The fixed 15:30 cutoff is a measurement convention. After 3 August it omits ten minutes of domestic trading, so 38.7 percent must not be interpreted as a pure overnight or offshore-only

share. For a short option position the consequence is sharper: a writer who collected a small premium for a strike some distance from the market can see the market reach that strike in the gap, with no opportunity to close in between. *Contextual Intelligence* (Gupta, 2026c) describes the same phenomenon from the side of events: an event's effect reaches different instruments at different times, and the participant who sees the chain first has time to act before the market does. Offshore index trading provides an additional venue for reacting to news. This sample does not measure its contribution to price discovery.

The outline of this paper proposed equalising offshore and domestic trading hours. That proposal is not carried forward. Longer domestic hours would extend the time in which individuals can trade, which on SEBI's evidence tends to increase losses. Section 10 proposes instead that overnight risk be stated at the point of trade.

7. Expiry mechanics: what is believed and what is documented

7.1 Two popular explanations

Two explanations of retail losses on expiry days circulate widely in Indian trading forums and channels, and the outline of this paper adopted both. The first is that option sellers steer the index toward the "max pain" strike, the settlement level at which option holders in aggregate receive least. The second is that on the morning after each weekly expiry, foreign participants rolling their hedges create a mechanical bid that lifts the index in the first minutes and then fades. The first has a serious academic ancestor in the pinning literature (Section 2.2). That a mechanism exists elsewhere is a reason to test for it here, not to assume it.

7.2 Max pain

The test uses 19 explicitly identified expiry series in official NSE and BSE daily bhavcopies. The dates are the Tuesday NIFTY and Thursday SENSEX expiries in two fixed windows, 24 June to 23 July and 16 September to 22 October 2026, with the immediately preceding session available. The holiday-shifted NIFTY expiry of 19 October is outside this weekday screen. This is a small, selected sample, not every expiry in June to October.

For each expiry, the previous session's open interest is filtered by the exact expiry date. Max pain is the strike that minimises the total intrinsic payout to holders of those options. Its absolute distance from the expiry-day underlying close is compared with the distance from the previous session's underlying close.

Index	Period	Expiries	Max pain closer	Median max-pain miss (points)	Median previous-close miss (points)
NIFTY	Jun to Jul 2026	4	3	73.3	65.7
NIFTY	Sep to Oct 2026	4	2	230.0	152.8
SENSEX	Jun to Jul 2026	5	3	158.2	238.2
SENSEX	Sep to Oct 2026	6	4	848.5	808.0
All		19	12		

Table 4. Explicit-expiry max-pain reconstruction, split at 3 August 2026. Source: official NSE/BSE daily bhavcopies; Appendix A.4-A.5. A smaller miss is better.

Max pain was closer in 12 of 19 observations. Its median miss was larger in three of the four groups, and smaller for SENSEX in June to July. The win count and miss magnitudes measure different things. With only 19 selected observations, these results do not establish a reliable forecasting advantage, nor do they rule one out. They do not establish that option sellers steer settlement. The test's selection of ordinary weekday expiries and its use of an underlying close rather than a separately reconciled final settlement file are further limitations.

The original draft used a lowest-price proxy because the author-held option chain lacked expiry identifiers. That proxy can mix expiries. The results above replace that calculation with explicit exchange expiry identifiers; the original 10-of-19 result is not retained as evidence.

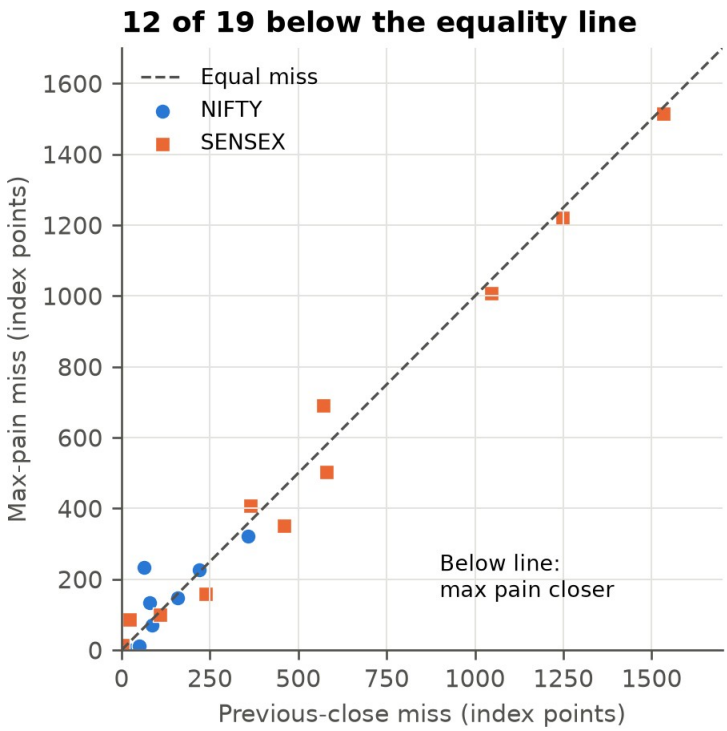


Figure 5. Max-pain forecast versus the previous-close forecast across 19 selected expiries. Points below the equality line mean max pain was closer; Table 4 reports 12 of 19. Distances are in each index's points, so NIFTY and SENSEX magnitudes are not directly comparable. The plot does not identify price manipulation.

7.3 The morning after expiry

For the NIFTY near-month future, each session following a Tuesday weekly expiry was compared with all other sessions on the overnight gap, the first fifteen minutes (09:15 to 09:30) and the rest of the day (09:30 to the close).

Sessions	Count	First 15 min up	Rest of day down	Median gap (bp)	Median first 15 min (bp)	Median rest of day (bp)
Day after Tuesday expiry	9	3	6	-23.7	-6.9	-27.8
All other sessions	46	27	21	+3.0	+5.6	+9.8

Table 5. Morning-after-expiry test. Source: author's tick record of the NIFTY near-month future, June to October 2026.

A mechanical bid after expiry would show as a rise in the first fifteen minutes in most post-expiry sessions. It showed in three of nine. On ordinary days the first fifteen minutes rose 59

percent of the time (27 of 46), so three rises in nine post-expiry mornings is below the ordinary rate, not above it.

The session the outline relied on, 15 July 2026, is one of the three, and it is striking (Figure 6). The previous day was a weekly NIFTY expiry; the July monthly contract did not expire until 28 July. The July future, which closed near 24,022 on 14 July, opened at 24,072, traded at 24,196 by 09:30 and reached 24,228, then fell to about 24,000 by 12:45 and closed near 24,072. A buyer at 09:30 was about 200 points down by early afternoon. The data used here does not identify a cause for the morning rise or the midday fall, and the paper does not attribute one. It reports the session because it is the kind of session that becomes a signal: one vivid instance, told afterwards as a rule. With nine observations the paper can say only that a systematic morning bid is not present in this sample, not that it never occurs.

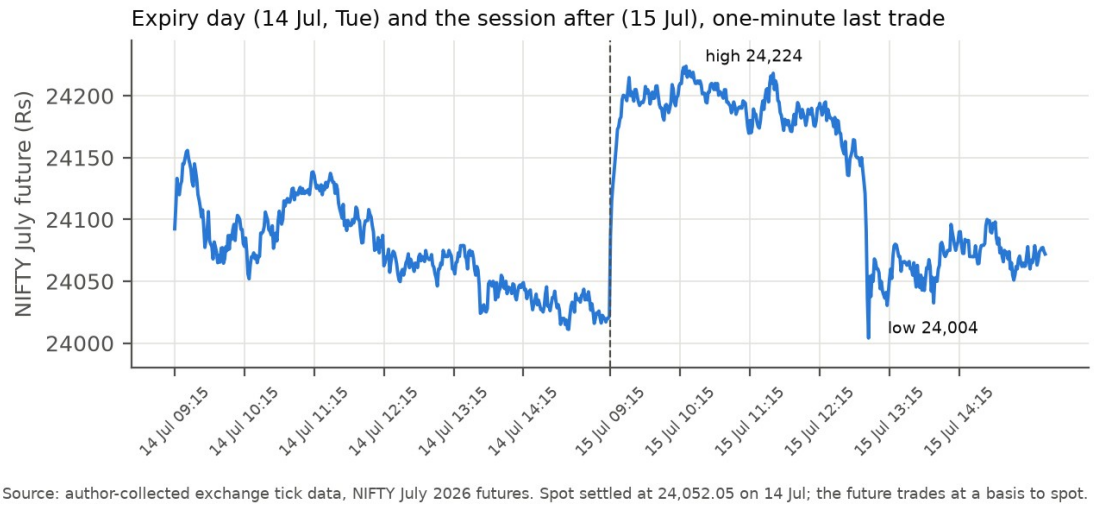


Figure 6. NIFTY July 2026 future, one-minute last trade, 14 July (weekly expiry) and 15 July 2026. Source: author-collected exchange tick data.

7.4 What SEBI has documented

The regulator has documented something more specific than either popular story, in two interim orders. Both set out prima facie findings at an interim stage; both are contested; neither is a final determination.

The intraday pattern (SEBI, 2025g). SEBI's interim order of 3 July 2025 concerning the Jane Street group examined January 2023 to March 2025, in which the group's index options profit was Rs 43,289.33 crore. The order explains why it looked at expiry days: the group's activity was "heavily concentrated around expiry days", and on such days "the index closing value has direct bearing on options payoff, thereby creating both the motive and the opportunity for influencing settlement outcomes". It selected the most profitable day, 17 January 2024 (Rs 734.93 crore), for detailed analysis and described what it called an "Intra-day Index Manipulation" strategy, which it found on 15 of 18 BANKNIFTY days examined:

- *Patch I, 09:15 to about 11:47.* BANKNIFTY had opened sharply lower, at 46,573.95 against a previous close of 48,125.10. In what the order calls "an otherwise falling

market", the group made net purchases of Rs 4,370.03 crore of BANKNIFTY constituents in the cash and stock-futures markets, which the order says supported or pushed up the constituents and the index. In the same patch it sold at-the-money BANKNIFTY calls and bought at-the-money puts. Its delta position in index options went from minus Rs 7,311.19 crore at the open to minus Rs 39,426.15 crore by 11:47, an increase in the short position of Rs 32,114.96 crore, which the order notes was 7.3 times the long position built in the constituents.

- *Patch II, 11:49 to 15:30.* The group sold the constituents it had bought and also sold BANKNIFTY futures, "reversing the aggressive purchases of patch I". The index closed at 46,064.45, below its open and near the day's low of 45,979.60.

Figure 7 sets out the sequence as the order describes it.

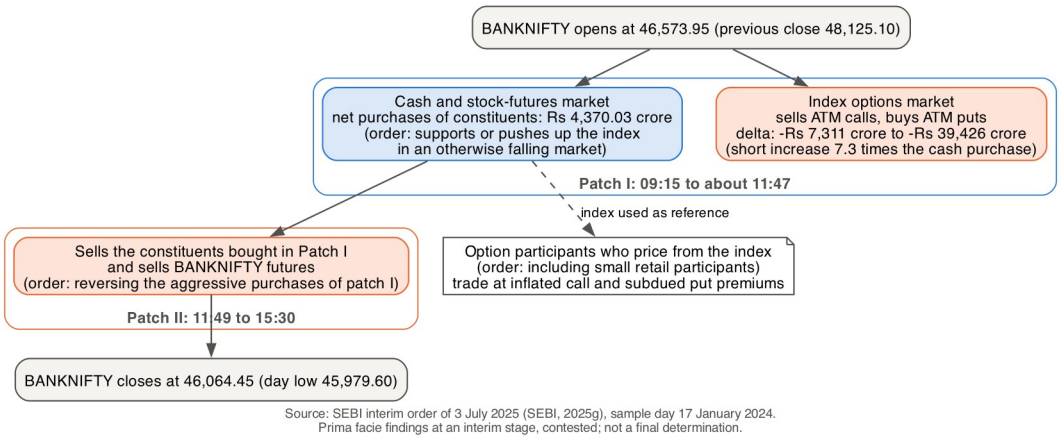


Figure 7. The two-patch pattern described in SEBI's interim order of 3 July 2025, sample day 17 January 2024. Prima facie findings at an interim stage, contested by the parties. Source: SEBI (2025g).

The order's account of who bore the cost is the part that matters for this paper. It observes that there are many more participants, "including small retail participants", in BANKNIFTY options than in the underlying markets; that "the vast majority of BANKNIFTY options participants that do not participate in the underlying cash or futures markets look to the index, determined by the cash market levels, to price and take a view on index options"; and that the Patch I buying "created conditions that misled BANKNIFTY index option participants with artificially inflated call option premiums and subdued put option premiums", allowing the group to enter its options positions "at very favourable pricing". The order described the Rs 199.7 crore lost on the cash and futures legs over the 15 days as "a mala fide cost". It also described an "Extended Marking The Close" strategy on three further BANKNIFTY days and three NIFTY days in May 2025, and impounded Rs 4,843.57 crore. The amount was deposited in escrow in July 2025 and some directions ceased to apply; the cease-and-desist direction continues pending investigation (SEBI, 2025h, 2025i). This account is limited to the cited interim order and July 2025 releases; it does not certify the subsequent procedural status.

The settlement auction (SEBI, 2026e). SEBI's interim order of 19 August 2026 concerned the SENSEX weekly expiry of 13 August 2026, ten days after the closing auction began to set the

expiry settlement price. The SENSEX reference price at 15:15 was 77,829.60. During the auction the order records three spikes in the SENSEX indicative price: from 77,661.40 to 78,023.42 (362 points) within two seconds, a further 133 points within 12 seconds, and 405 points within 28 seconds. It examined the trading "having regard to ... the concurrent F&O positions of the concerned participants" and impounded Rs 3,67,80,773 from two noticees. Eight weeks later SEBI reverted the settlement method (SEBI, 2026d).

What the two orders establish for this paper. Neither describes settlement being drawn to max pain, and neither describes a mechanical morning bid. Both describe the same structure from different sides: the price an option settles or trades at is set by a smaller market (index constituents, or a closing auction) than the option market itself, so a participant with a large derivatives position and the capacity to trade in the smaller market can, on the regulator's prima facie findings, move the reference price that many other option holders rely on. The intraday pattern worked, on the order's account, because option participants priced from the index and could not see who was moving it or why. That is a provenance problem in a second form: the individual sees a price, as they see a signal, and has no record of what caused it.

The distinction matters for policy. A mechanism documented in a regulator's order can be addressed by surveillance and rules, and SEBI has acted: intraday limits in September 2025, a settlement change in October 2026. A mechanism that is only believed, and repeated in channels that sell signals, cannot be addressed by anyone. Its main effect is to give individual traders a confident account of why they lost that leads them to trade the next expiry.

8. Provenance: the unifying gap

8.1 What Part 1 established

A retail trader's decision to buy a call on expiry morning rarely begins with the trader. It begins with a message: a Telegram channel's call, a screenshot of an option-chain "analyser", a video about max pain, or the output of an AI tool. *Signal Provenance in Retail Algorithmic Trading* (Gupta, 2026a) showed that this upstream layer is outside every regulatory framework it examined. The broker's order interface receives symbol, quantity, price and side; it never receives what caused the order. SEBI's framework of 4 February 2025 tags each algorithmic order, makes the broker principal, empanels the provider and fixes the IP address, and so makes the order chain fully traceable. What caused the order stays outside it. Enforcement against unregistered recommendations proceeds by reconstruction: in the four SEBI matters that paper examined, after a complaint, by forensic audit of messages and bank trails, between roughly two and a half and six years after the conduct began. Part 1 also showed, in production, that the origin of a signal can be captured on the device where it is received, before any order exists, and proposed a diagnostic: for any framework, ask what the regulated entity actually receives, and whether the conduct of concern is present in it.

8.2 The same absence, wherever the order chain is regulated

SEBI's own framework states what it records, and its operative clauses are worth quoting rather than summarising. Under the circular of 4 February 2025 (SEBI, 2025a), "brokers shall be the principal while any algo provider or fintech/vendor ... shall act as its agent" (paragraph I(a)); "All

algo orders originating/flowing through Application Programming Interface (API) extended by brokers to algo providers, shall be tagged with a unique identifier provided by Stock Exchange" (paragraph I(b)), "in order to establish audit trail" (paragraph II(b)); access is only "through a unique vendor client specific API key and static IP whitelisted by the broker to ensure identification and traceability of the algo provider and the end user (i.e. investor)" (paragraph I); and the provider of an algorithm whose logic is not disclosed must "Register as a Research Analyst and maintain a detailed research report for each such algo" (paragraph V). Every clause identifies a party: the broker, the algorithm, the provider, the investor. None records an event before the order: the message, recommendation or output that prompted it. The framework achieves exactly what it says, the "identification and traceability" of who placed an order and through what. Its audit trail begins at the order.

Part 1 read MiFID II Article 17 and its implementing standard, RTS 6, in full, and found that the word "signal" does not appear in either; the European framework requires a firm to know which algorithm, trader or client is responsible for an order, not what caused it. The United States' Consolidated Audit Trail, the most detailed order-level record any regulator has built, confirms the pattern from a third jurisdiction. Rule 613 requires that, for the original receipt or origination of an order, the reporting firm record the Customer-ID of each customer, an order identifier, the identifier of the broker-dealer receiving or originating the order, the date and time, and the material terms of the order (17 CFR 242.613(c)(7)(i)). It reaches the customer. It does not reach the cause. Every framework examined anchors at the order, because the order is where an accountable party first appears, and every one is silent on what came before it (Figure 8).

8.3 Why provenance connects the other asymmetries

Set the four asymmetries of this paper against the diagnostic of Part 1.

- *Cost floor and counterparties (Sections 5.1, 5.2).* SEBI publishes them, annually, in a study. The regulator knows; the individual placing an order does not see it.
- *Positioning (Section 5.3).* NSE publishes it, daily, in a CSV. The regulator and any analyst can see it; the individual at the order screen does not.
- *Time (Section 6).* The overnight market is public and its effect is measurable. It is not stated where the decision to carry a position is made.
- *Provenance (Section 8).* Nobody records it. The regulator cannot see it, the broker is never sent it, and the individual usually cannot verify it.

The first three are display problems: information that exists and is shown in the wrong place. The fourth is a capture problem: information that exists for a moment, on a phone, and is then lost. The display problems are why individuals keep taking the losing side without knowing it is the losing side. The capture problem is why the regulator, which can see the aggregate result, cannot see its cause, and so regulates the visible artefact, access, instead.

The two channels of loss in Section 5.2 are not equally exposed to the capture problem. Provenance would be most immediately useful in the first channel, the intraday buying of short-dated options, where a trade is opened and closed within one session and the prompt for it is most likely to have arrived on a phone that morning. The second channel, carried positions on the other side of foreign hedges, is better addressed first by the positioning disclosure of Proposal 1.

The scale of what the regulator cannot attribute can be stated without guessing. Individuals' gross trading loss in FY26 was about Rs 72,000 crore (SEBI, 2026a). Because no framework records what prompted any order, the share of that loss that followed registered advice, the share that followed an algorithm, and the share that followed an unregistered channel or an AI tool are all unknown, to the regulator and to everyone else. The paper does not estimate those shares; any percentage offered here would be invented, and an invented percentage would be quoted as a finding. The public data used here cannot allocate those losses by signal source. A reliably populated provenance field could permit an aggregate comparison, subject to validation and linkage to outcomes.

Three findings of this paper show the connection concretely. First, the claims tested in Section 7 are exactly the kind of content that circulates as signals. A trader who believes settlement is steered toward max pain has a reason to trade every expiry, and a channel that teaches the belief has a reason to sell the next signal; nothing in the order chain records that the belief caused the trade. Second, positioning is collective. The Client category's crowding into one side of the index book, and the concentration of individual option buying on expiry days, are aggregate outcomes of many decisions that share sources: courses, channels, tools. Without provenance, neither the regulator nor the trader can see how much of a crowded position was caused by a small number of upstream sources. Third, AI tools now generate signals at scale. *Beyond the Mirror* (Gupta, 2025) found that conversational models drift from their own safeguards under sustained, polite framing; a trader who asks an AI assistant whether to buy a call before expiry is not dealing with a registered research analyst, and nothing records that the prompt for the trade came from a model.

The documented cases in Section 7.4 add a fourth connection. The intraday pattern described in the Jane Street order worked on participants who took their cue from a reference price without knowing who was moving it. A signal is a reference of the same kind. In both cases the individual acts on information whose origin is invisible to them, and in both the regulator reconstructs the origin afterwards.

8.4 The regulatory record

SEBI's order of 30 April 2026 against the Telegram channels "Intraday Jackpot" and "Professional Day Trading Institute" records about Rs 9.02 crore collected from subscribers without registration (SEBI, 2026f). The October 2024 association rule and January 2025 clarification constrain regulated entities' links to unregistered advice providers (SEBI, 2024c, 2025k). They do not prescribe a category of signal origin on each order.

AI use is not unregulated. Regulation 16C of the SEBI Intermediaries Regulations, inserted with effect from 10 February 2025, assigns regulated persons responsibility for investor data, AI outputs and legal compliance (SEBI, 2025j). The narrower gap argued here is that this provision does not create a signal-to-order provenance field. MeitY's November 2025 guidelines work through existing law (MeitY, 2025). Its February 2026 IT Rules amendment addresses synthetically generated audio, visual and audio-visual information (MeitY, 2026). Labelling that content does not link it to a resulting trade. Figure 8 illustrates the proposed additional category; its reliability remains to be tested.

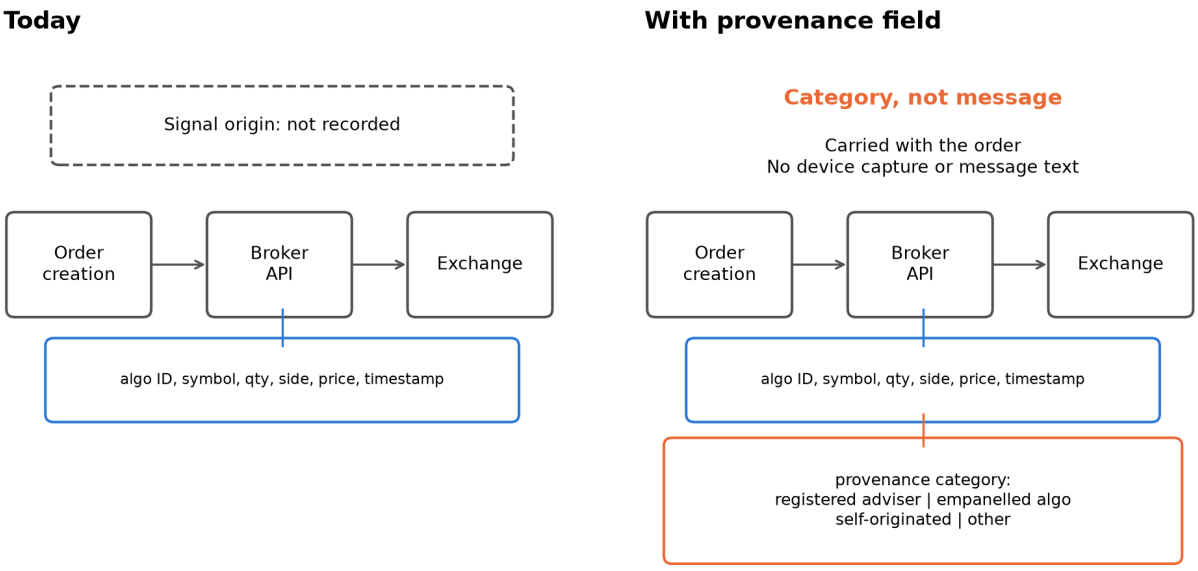


Figure 8. The order chain records what was traded, not what caused the trade. A provenance category is proposed as a small field for observing the reported source category in aggregate. It carries no message content and does not establish causation or verify the declaration. Sources: SEBI (2025a); author's proposal.

8.5 What a provenance field would be

Part 1 demonstrated capture at the device. A regulator does not need to begin there. The smallest step that would change what the order chain records is a provenance field: one more attribute carried with each order, alongside the algo identifier that SEBI's framework already requires.

- *Content.* A category, not a message: whether the order was prompted by registered advice, by an empanelled algorithm, by the trader's own analysis, or by another source. It would carry no message text and no screenshot. How a category is established and verified is a design question for the regulator and the exchanges, and this paper does not specify it.
- *Point of attachment.* The order, at the point where the algo identifier that SEBI's framework already requires is attached.
- *Carriage and retention.* Carried with the order to the exchange like the algo identifier, and retained with the order record under the existing record-keeping obligations of the broker and exchange. Access would be under the regulator's existing powers, not a new channel.
- *Use.* Two uses, one aggregate and one forensic. In aggregate, the exchanges could publish the share of individual index options turnover, and of expiry-day turnover, by provenance category; for the first time the regulator would know how much of the losing flow is prompted by registered advice, by algorithms, and by sources outside the regulated perimeter. In enforcement, a complaint about a channel would start from orders already marked as prompted by an unregistered source, rather than from screenshots.

The limits are the ones Part 1 stated, and they are real. A field is only as reliable as the means by which it is filled, and a determined actor can route around any capture point. Capturing anything about what a person received on a private device raises privacy questions that a regulator must settle in public before any mandate: what is retained, for how long, visible to whom, revocable how. The field is proposed because the category is narrower than content and the aggregate use needs no identity at all, not because those questions are answered. Its purpose is to end the ambient invisibility in which unregistered channels currently operate at scale, not to defeat every evasion. And no claim is made that provenance reduces losses; Part 1 made no such claim and neither does this paper. The claim is that without it, the regulator can see that individual traders are crowded into one side of the market and onto one day of the week, and cannot see who put them there.

8.6 Piloting the field without a mandate

Because the field is untested, the sensible first step is a pilot, not a rule. A pilot could run under SEBI's regulatory sandbox or by agreement between one exchange and a small number of brokers, with no enforcement attached:

- *Scope:* individual clients' orders in one index options contract (for example the weekly NIFTY contract), for a fixed period of several expiry cycles.
- *Content:* the categories of Section 8.5 only; no message content and no personal identity.
- *Output:* the participating exchange publishes, after the pilot, the share of pilot orders and turnover in each category, the share left blank, and the expiry-day share in each category.
- *Questions the pilot answers:* how often the field is filled at all; whether the categories are stable enough to be useful; whether the aggregate shares differ by time of day or expiry day; and what privacy questions arise in practice, so that they can be settled in public before any wider rule is considered.

A pilot of this kind could fail, by showing that the field is left blank or filled carelessly. That would itself be a finding about what any workable method must provide. Either outcome tells the regulator something it does not know today.

9. Two months of exit: August to October 2026

9.1 The flows

Between 21 August and 22 October 2026, 42 trading sessions, foreign institutional and portfolio investors sold a net Rs 1,06,956 crore in the Indian cash market and domestic institutional investors bought a net Rs 1,57,928 crore, according to NSE's provisional daily data (Figure 9). Foreign investors were net sellers on 32 of the 42 sessions, including 15 consecutive sessions from 24 September to 15 October. The selling came in two phases: a slow drain to the third week of September, then an acute month.

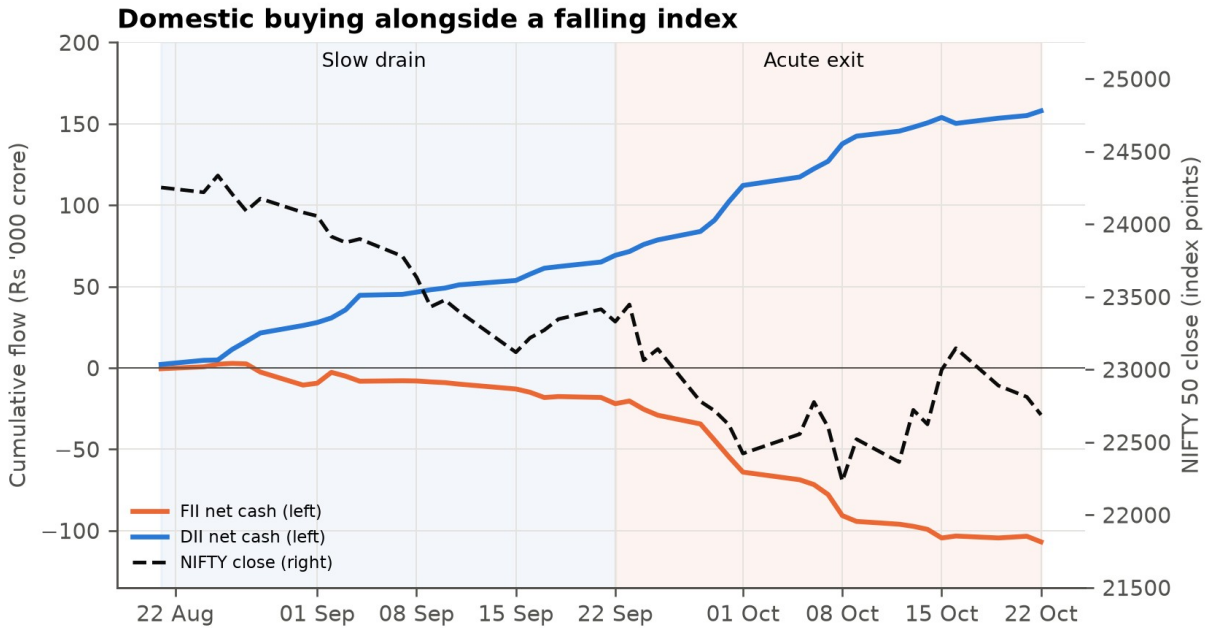


Figure 9. Cumulative FII/FPI and DII net cash flows with NIFTY close, 21 August to 22 October 2026. Flows are from NSE provisional data, with nine sessions from a secondary reprint; closes are NSE data. The chart shows domestic net buying alongside the fall; it does not establish a causal effect or test analysts' explanations. The two axes have different units.

Period	Sessions	FII/FPI net (Rs crore)	DII net (Rs crore)	DII buying as share of FII selling
21 to 31 August 2026	7	-10,589	+26,023	246%
September 2026	21	-44,013	+76,030	173%
1 to 22 October 2026	14	-52,354	+55,875	107%
21 August to 22 October 2026	**42**	**-1,06,956**	**+1,57,928**	**148%**
Phase 1: 21 August to 21 September	21	-18,206	+65,039	357%
Phase 2: 22 September to 22 October	21	-88,750	+92,889	105%

Table 6. NSE provisional cash-market flows. The first three rows sum to the window total; the two phase rows are an alternative split of the same 42 sessions and also sum to it. Source: NSE FII/DII trading activity, the author's daily collection, with nine sessions taken from Trendlyne's reprint of the NSE series; totals computed by the author and cross-checked. For reference, full-month August 2026 was -7,532 (FII) and +58,268 (DII).

The largest single-day foreign outflows were on 8 October (Rs 12,943.58 crore), 30 September (Rs 10,148.41 crore), 29 September (Rs 9,980.22 crore) and 1 October (Rs 9,484.22 crore). On those four days domestic institutions bought between 70 and 111 percent of the amount sold.

The depository series, which includes primary-market purchases, tells the same story with different numbers. Foreign portfolio investors were net buyers of Rs 20,200 crore in July and Rs 29,631 crore in August, then net sellers of Rs 35,861 crore in September (secondary market -45,537, primary market +9,676) and of Rs 48,211 crore in October to the 22nd (News On AIR, 2026; Free Press Journal, 2026; Trendlyne, 2026). The calendar-year 2026 total to 22 October was a net outflow of about Rs 3.09 lakh crore, against about Rs 1.66 lakh crore in the whole of 2025. The two series should not be mixed: in August the depository series was positive and the exchange series negative, because of primary-market buying, reporting lags and coverage.

Market analysts at the time gave consistent reasons: US ten-year yields above 5.2 percent, a stronger dollar, crude oil near 100 dollars a barrel amid the West Asia war, and a reallocation of global capital toward AI-linked markets in North Asia. One called it "global capital repositioning" rather than weakness in India's economy (Free Press Journal, 2026). These are the analysts' explanations, quoted from the press; this paper does not test them. The macro context is the one *Asymmetric Multipolarity* described, in which financial centrality and real-economy capacity have decoupled and portfolio capital can leave a growing economy quickly (Gupta, 2026f). The rupee, under pressure earlier in the year (Gupta, 2026b), ended the window slightly stronger, at 94.82 per dollar against 95.71.

9.2 What domestic absorption did

Domestic institutions absorbed the cash-market exit and more. Their buying was funded in large part by monthly SIP contributions, which reached records of Rs 32,297 crore in August and Rs 32,490 crore in September, and by equity mutual fund net inflows of about Rs 29,329 crore and Rs 38,307 crore in those months (as reported from AMFI data; Angel One, 2026; Kotak Neo, 2026). This is the absorption *The Great Indian Illusion* documented in the first half of 2026 (Gupta, 2026b), continuing at scale.

Domestic buying coincided with a fall. The data does not estimate how prices would have moved without it. NIFTY 50 closed at 24,252.00 on 21 August and 22,687.65 on 22 October, a fall of 6.45 percent, with its lowest close, 22,231.80, on 8 October, the day of the largest outflow (NSE daily index data). SENSEX fell 5.53 percent, from 77,540.83 to 73,249.39. A brokerage note at the time put it accurately: institutional demand "can soften the impact of selling, but it cannot fully insulate the market" (5paisa, 2026).

The deeper point is that absorption operates in the cash market only. A domestic mutual fund that buys the shares a foreign investor sells is a cash buyer of shares. It does not take the other side of the foreign investor's index hedge. That side, as Section 5.3 showed, belongs mostly to the Client category.

9.3 What the exit cost the retail side of the derivatives market

Public data permits an illustrative calculation for index futures; it does not bound the rest of the portfolio.

For index futures, each category's daily mark-to-market is its net position at the previous close multiplied by the change in the index and the contract multiplier. Using NSE's participant file for positions, NSE's daily NIFTY 50 closing values for prices, and the NIFTY lot size of 65 in force since January 2026 (NSE, 2025), and treating every index future as NIFTY-equivalent, the Client category's index-futures position lost an estimated Rs 2,119 crore over the window. The FII category's net short position gained an estimated Rs 2,546 crore; Pro and DII positions lost Rs 289 crore and Rs 138 crore. The four figures sum to zero, as they must. The NIFTY-equivalent treatment is an approximation: the participant file aggregates all NSE index futures, and the other contracts (BANKNIFTY at a lot of 30, for example) have notional values of the same order but did not move exactly with NIFTY.

That is the futures book alone, and futures were the smaller part of the Client category's exposure. Over the same window the Client category was net short about 742,000 index put contracts. When the index falls 6.45 percent in two months, with several daily moves above one percent, short puts lose more than in proportion as strikes move toward the money. This paper does not estimate that loss, because the participant file does not report the strikes at which puts were written, and an estimate without them would be invented. Nor does it estimate the intraday option-buying losses of the first channel (Section 5.2), which do not appear in end-of-day open interest. The futures figure is an illustrative futures-only mark-to-market. It is neither realised profit and loss nor a lower bound on total derivatives losses: unobserved options and other positions could offset it.

For scale, SEBI's figure for individuals' net losses in FY26 was Rs 91,685 crore on its broker sample, about Rs 7,600 crore a month on average, in a year without a comparable foreign exit (SEBI, 2026a). It is a reference point, not an estimate for this window.

The episode shows the structure in two months. Foreign investors sold Indian shares and held index hedges that paid them as the market fell. Domestic institutions, funded by households' monthly savings, bought the shares. The Client category, which contains individual traders but has no verified individual allocation, held the other side of the hedges. Household savings and potential trading exposure are two distinct channels; their overlap cannot be measured here. The first role is protected by the mutual fund rules. The second is the one in which, by the regulator's count, close to nine in ten lose. Figure 10 draws the two roles.

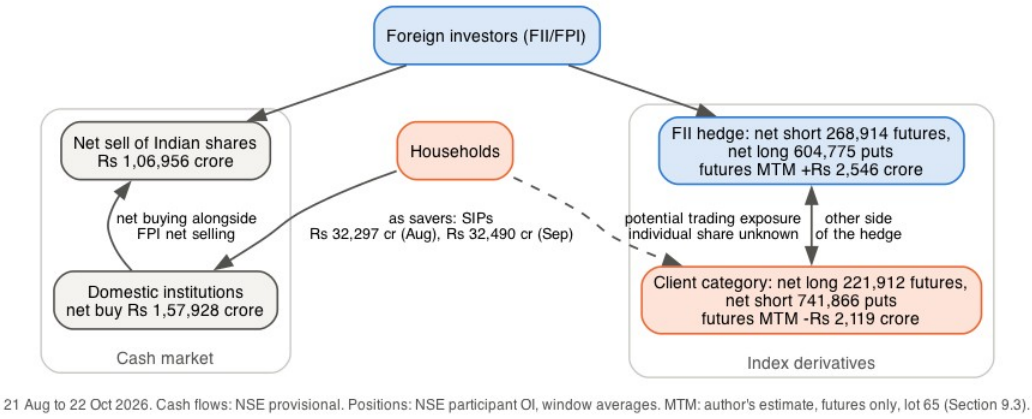


Figure 10. Two potential household exposures in the August to October 2026 exit. The dashed link to Client positioning is a hypothesis, not an estimated individual allocation. Institutional flows do not identify which investor bought a particular seller's shares. Sources: NSE provisional FII/DII data; NSE participant-wise open interest (window averages); AMFI data as reported; author's futures mark-to-market estimate (Section 9.3).

10. Proposals: from access control to structural transparency

What a regulator could do tomorrow. The most immediately actionable step needs no new data, no new legislation and no new capture of anything. NSE already publishes the participant-wise open interest file every evening, and SEBI already publishes the category-wise profit and loss of the segment. A SEBI direction to brokers, under the existing risk-disclosure framework, to show a plain-language line from each at the point where an individual places an index derivatives order would put information institutions already use in front of the person who is missing it. That is Proposal 1. The other proposals need more: an exchange computation (Proposal 2), a pilot and then possibly a new order attribute (Proposal 3), and exchange publication of category data that is not yet published (Proposal 4).

Each proposal follows from the evidence and is stated with its scope, its enforcement and its limit. None bans individuals from derivatives, raises their costs or restricts their access further.

Proposal 1. Show positioning and counterparties where the decision is made. The previous day's participant positioning in the instrument, in one plain line (for example: "Yesterday, client accounts held 83 percent of their index futures long; foreign institutions held 90 percent short"), and SEBI's latest category-wise result in one sentence (for example: "In FY26, individuals lost about Rs 72,000 crore before costs in equity derivatives; algorithmic proprietary and foreign traders earned most of the profits"), shown on the order screen for any index derivative.

- *Scope:* order entry for index derivatives by individual clients, on every broker interface.
- *Source and accuracy:* the line is generated by the exchange from the participant file it already publishes, and distributed as a standard daily text through the exchange's data

feed, so no broker computes or edits it; the category sentence is set by SEBI when each study is published.

- *Enforcement:* a SEBI direction to stock brokers under the existing risk-disclosure framework, verified in the exchanges' existing broker inspections.
- *Limit:* disclosure changes behaviour modestly, and a participant determined to trade will trade.

The outline of this paper proposed real-time publication of aggregate dealer gamma exposure, by analogy with the United States. That proposal is withdrawn as stated: the author could not verify that any US clearing house or exchange publishes dealer gamma exposure as such, and the Indian exchanges and SEBI already publish more participant detail than the proposal assumed. The gap is not in what is published but in where it is shown.

Proposal 2. State overnight risk as risk.

- *Scope:* any index derivative position held by an individual client within the last hour of trading.
- *Content:* the recent distribution of overnight gaps for that instrument, computed and published by the exchange, and a sentence in the risk disclosure document stating that Indian index futures trade offshore outside domestic hours, using dated exchange hours rather than the fixed sample cutoff.
- *Enforcement:* as for Proposal 1; the risk disclosure document is amended by SEBI.
- *Limit:* this informs; it does not protect.

Proposal 3. Make signal origin observable: the provenance field of Section 8.5.

- *Scope:* all orders in equity derivatives from individual clients, beginning with index options.
- *Attachment:* at order creation, alongside the existing algo identifier; the method of establishing the category is set by the regulator after the pilot of Section 8.6.
- *Retention and access:* with the order record under existing obligations; regulator access under existing powers.
- *Publication:* the exchanges publish, monthly, the share of individual index options turnover and expiry-day turnover by provenance category.
- *Enforcement:* brokers are obligated to carry the field; registered research analysts, investment advisers and algo providers remain accountable under their existing regulations for any order attributed to them.
- *Limits:* those in Section 8.5, including privacy questions that must be settled in public before any mandate.

Proposal 4. Evaluate expiry design by who holds what at settlement. Expiry reforms have been justified by the volume of speculative trading on expiry days. The evidence here and the regulator's own orders suggest that the more important variables are which categories hold the expiring positions and who can move the reference price.

- *Scope:* index derivatives expiring on each exchange.

- *Measure:* the share of expiring open interest held by each participant category on each expiry day, and the distribution of expiry-day profit and loss by category, both of which the exchanges can compute.
- *Timing:* SEBI's change from closing-auction settlement to a 30-minute VWAP, effective 7 December 2026, is a natural experiment; the exchanges can publish both measures for the months before and after it.
- *Enforcement:* a SEBI direction to exchanges to publish.
- *Limit:* this paper does not show that any particular expiry design is better.

11. Limitations and falsification

The counterparty and cost evidence (Sections 3.2 and 5.1) is SEBI's own. The gross category figures are for the whole market and the cost figures for the broker sample, so they show the structure, not an exact ledger.

The positioning evidence (Section 5.3) rests on NSE's own data. It is limited by the breadth of the Client category and the absence of any published estimate of the individual share within it. If the exchanges were to publish that individuals hold a small share of Client index positioning, the inference from the category to individual traders would weaken; the first channel (Section 5.2) would not.

The time and morning-after evidence rests on the author's tick records, which have gaps. The max-pain test uses an explicit-expiry reconstruction from official exchange files over two selected windows (Appendix). The samples are short: 55 session pairs, 19 expiries, 9 post-expiry mornings, across a change in settlement method. The paper reports what they show and does not generalise beyond them. A longer, complete sample showing max pain outperforming the previous close, or a consistent morning bid after expiry, would change the evidence discussed in Sections 7.2 and 7.3.

The documented cases (Section 7.4) are interim orders with prima facie findings, contested by the parties. If they are set aside, the structural point that a smaller market sets the reference price for a larger one remains, but the evidence that it has been exploited would weaken.

The loss estimate (Section 9.3) treats all index futures as NIFTY-equivalent and covers futures only. It is an illustrative mark-to-market, not realised profit and loss or a lower bound on total portfolio losses.

The flow analysis (Section 9.1) uses NSE's provisional series; final figures can differ, and the depository series differs in coverage. Analysts' explanations of the exit are quoted, not tested. Several figures were taken from reprints of primary releases (NSDL, AMFI) that the author could not open directly; they are labelled as such in the references.

The provenance argument (Section 8) is an argument about what the regulator can and cannot see. No claim is made that provenance capture would reduce losses, and no data exists yet to test it.

The central claim would be falsified if the share of individual loss-makers continued to fall materially in FY27 and after, toward the share who lose before costs, while the counterparty and

positioning structure in Sections 3.2 and 5.3 persisted. The FY26 fall of 3.2 points is the first in SEBI's series. A sustained trend would weaken the paper's policy argument; identifying the effect of access rules would still require a causal design.

12. Conclusion

SEBI's measures since 2024 have been careful, consulted on and phased, and they have done what they were designed to do: they have changed contract sizes and margins. Participation and expiry-day turnover shares subsequently fell, though these comparisons do not identify the effect of the rules. SEBI's surveillance has also found and acted on specific conduct at expiry. The measures were not designed to change the cost floor, who sits on the other side of the individual's trade, the side individuals end up carrying, the hours in which that side's risk moves, or the invisibility of what led them there. The regulator's studies motivate examining these features alongside access; they do not estimate each feature's causal contribution to losses.

The first paper in this series showed that the order chain cannot see what caused an order. This paper has argued that the same gap is why the other asymmetries persist: they are published, but never connected to a decision, and so never shown where the decision is made. The two months from August to October 2026 showed the structure in motion. Foreign investors left the cash market and were paid by their hedges; domestic institutions, funded by households' savings, bought what they sold; the Client category held the other side of the hedges.

The solution argued for here is not to keep individuals out of derivatives. It is to show them, at the point of decision, what institutions already see, who holds which side, who earns from it, and when the risk moves, and to begin recording where the signal came from.

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Appendix: data and methods

A.1 Public data. NSE participant-wise open interest (daily; categories Client, DII, FII, Pro; long and short positions separately, in number of contracts); NSE provisional FII/DII cash-market activity (daily); NSE daily index closing values; SEBI studies, circulars and orders; and the sources in the references.

A.2 Author-held data. (a) A daily end-of-day option chain for NIFTY, BANKNIFTY and SENSEX, 23 June to 22 October 2026, 47 trade dates, with a gap from late July to mid September; it carries strike, option type, open interest, change in open interest, last price and underlying price, but no expiry date. (b) An exchange tick record of NIFTY futures, with a gap from late July to 16 September 2026. (c) A daily copy of the NSE FII/DII provisional series from 27 April 2026, missing nine sessions in the window (4, 7, 14, 21, 25 and 27 August; 1, 9 and 16 September), which were taken from Trendlyne's reprint of the NSE series, rounded to one decimal. The author's copy matched published figures on the three dates checked (30 September, 1 October and 8 October 2026).

A.3 Short put share. For each daily file, the Client category's short index-put open interest (column "Option Index Put Short") is divided by the total short index-put open interest across the four categories; Table 3 reports the averages over the 41 files from 21 August to 21 October 2026 (3,246,595 of 4,646,222, or 69.9 percent). Net positions are long minus short for each category. The Client share of all index-option positions (62.9 percent) is the Client category's sum of long and short calls and puts divided by the same sum across categories.

A.4 Expiry identification and sample. Table 4 uses official NSE and BSE UDiFF bhavcopies, filtering symbol, CE/PE option type and exact XpryDt. The sample includes ordinary Tuesday NIFTY and Thursday SENSEX expiries within 24 June-23 July and 16 September-22 October 2026, with the preceding calendar day's exchange file. The 19 October holiday-shifted NIFTY expiry is excluded by this weekday screen. The retained data includes each observation and source-file hashes. The earlier lowest-price proxy is retired because it can select other expiries.

A.5 Max pain. For strikes K with open interest OI_c(K) and OI_p(K), max pain is the listed strike S minimising the sum of OI_c(K) x max(0, S - K) + OI_p(K) x max(0, K - S), using the previous session's exact expiry series. The benchmark and outcome are the respective daily file's underlying prices. Equal misses are not wins. Medians are rounded to one decimal using decimal half-up rounding. Separate official final settlement prices have not been reconciled, so the result is stated as distance from the expiry-day underlying close. The selected observations are listed in Appendix A.11.

A.6 Overnight share. For consecutive sessions t-1 and t of the near-month NIFTY future with the same contract, gap(t) = |O(t) - C(t-1)| / C(t-1) and intraday(t) = |C(t) - O(t)| / O(t), where O is the first trade at or after 09:15 and C the last trade before 15:30. Overnight share = sum of gap(t) / (sum of gap(t) + sum of intraday(t)).

A.7 Morning after expiry. The near-month NIFTY future is the future with the earliest expiry on or after the session date. The gap runs from the previous session's last trade to the first trade at or after 09:15; the first fifteen minutes from that trade to the last trade before 09:30; the rest of the day from that price to the last trade before 15:30. Sessions on which the near-month contract changed are excluded.

A.8 Index-futures mark-to-market. For each category, MTM = sum over sessions t of N(t-1) x L x (P(t) - P(t-1)), where N is net index-futures contracts in the participant file, P the NIFTY 50 close and L = 65. The 22 October participant file was not yet published when the calculation was made, so 21 October positions are used for the final session.

A.9 Limitations of the NSE participant-wise open interest file. (a) It is an end-of-day snapshot: it shows positions carried past the close, not intraday positions, so same-day option buying and selling, the first channel of Section 5.2, does not appear in it. (b) It has four broad categories, and the author found no NSE definition of the Client category or estimate of the individual share within it (Section 5.3). (c) It reports contracts, not notional value, and aggregates all index futures and all index options across underlyings, so it cannot separate NIFTY from BANKNIFTY or other indices. (d) It does not report strikes or expiries, so the loss on the Client category's short puts cannot be estimated from it (Section 9.3). The files used are the daily files from 21 September 2020 to 21 October 2026 (1,502 per category); Table 3 and the window figures use the 41 files from 21 August to 21 October 2026; the 22 October file had not been published when the calculations were made.

A.10 Reproducibility. Queries, scripts, exchange-file hashes and the author-held records are retained and available to a regulator or reviewer on request. The explicit-expiry observations below make Table 4 independently recomputable.

A.11 Explicit-expiry observations. Distances are index points. Source: official NSE and BSE UDiFF bhavcopies for each listed expiry and its preceding session.

Index	Expiry	Previous close	Max pain	Expiry close	MP miss	Previous-close miss
NIFTY	2026-06-30	23946.25	24000.0	23865.75	134.25	80.50
NIFTY	2026-07-07	24430.35	24400.0	24398.7	1.3	31.65
NIFTY	2026-07-14	24211.0	24200.0	24052.05	147.95	158.95
NIFTY	2026-07-21	24238.5	24200.0	24187.7	12.3	50.8
NIFTY	2026-09-22	23414.3	23400.0	23329.0	71.0	85.3
NIFTY	2026-09-29	22780.25	22950.0	22716.2	233.8	64.05
NIFTY	2026-10-06	22555.75	22550.0	22776.1	226.1	220.35
NIFTY	2026-10-13	22363.35	22400.0	22721.95	321.95	358.60
SENSEX	2026-06-25	76991.22	77000.0	77100.47	100.47	109.25
SENSEX	2026-07-02	76922.64	77000.0	77502.12	502.12	579.48
SENSEX	2026-07-09	76503.6	76900.0	76741.82	158.18	238.22
SENSEX	2026-07-16	77185.43	77200.0	77186.87	13.13	1.44
SENSEX	2026-07-23	76755.05	76800.0	76391.39	408.61	363.66
SENSEX	2026-09-17	74336.45	74400.0	74314.59	85.41	21.86
SENSEX	2026-09-24	74828.25	74800.0	73580.54	1219.46	1247.71
SENSEX	2026-10-01	72480.29	72600.0	71909.7	690.3	570.59
SENSEX	2026-10-08	72638.7	72600.0	71593.24	1006.76	1045.46
SENSEX	2026-10-15	72781.89	72800.0	74315.61	1515.61	1533.72
SENSEX	2026-10-22	73709.69	73600.0	73249.39	350.61	460.30